W5 L1

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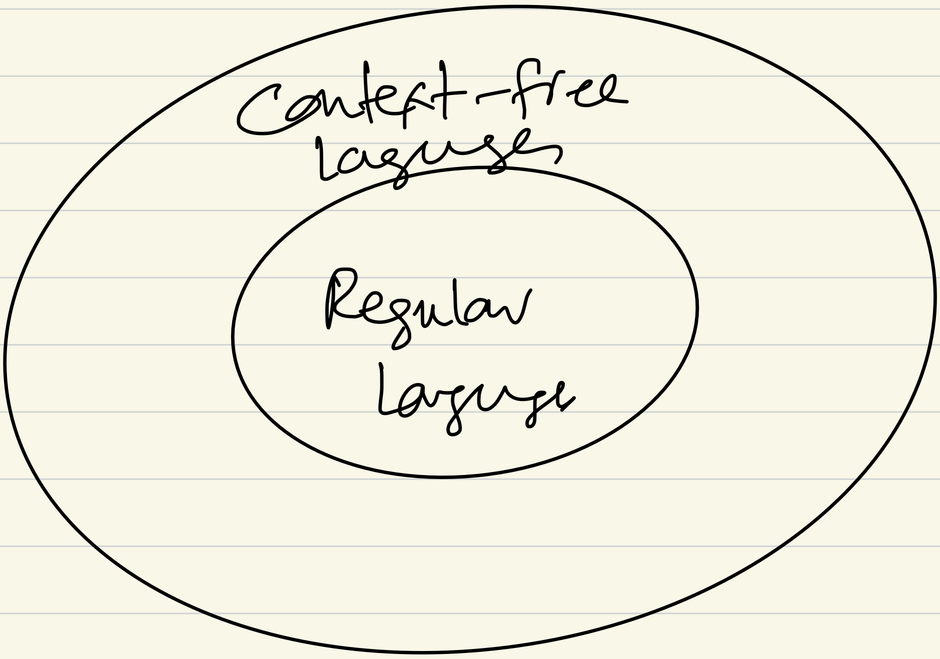
Now we know there are some languages which are not regular.

$\Sigma = \{0,1\}$

$\rightarrow L = \{0^n1^n \mid n \geq 0\}$



Chomsky Hierarchy

Grammar. Context-Free Grammar (CFG)

$\underline{A} \rightarrow \mathbf{0}A\mathbf{1}$ // substitution rule

$A \rightarrow B$ $\qquad \Sigma = \{0,1,\#\}$

$B \rightarrow \mathbf{\#}$

0, 1, #
Bold symbols mean "terminal symbol"
A, B are non-terminal symbols.

G_1 : $\Sigma = \{0, 1, \#\}$

$A \to 0A1$, $A \to B$, $B \to \#$

CFG : collection of such substitution rules

nonterminal symbol $\longrightarrow$ string of terminal & non-terminal symbols

Start symbol $= A$

we use grammar to describe a language.

A grammar G generates a language L. $L(G) =$

w, L, $w \in L$?

Derivation.

$A \Rightarrow 0A1 \Rightarrow 00A11$

$\Rightarrow 000A111 \Rightarrow 000B111$

$\Rightarrow 000\#111$

$A \Rightarrow B \Rightarrow \#$

$L(G_1) = \{\#, 0\#1, 00\#11, 000\#111, \dots\}$

$L(G_1) = \{0^n\#1^n \mid n \geq 0\}$

Definition: A CFG is 4-tuple

(V, Σ, R, S) where

- V: is a finite set of variables (non-terminal)
- Σ: alphabet: fixed & finite set of characters. (terminal symbols)
- R: set of rules //substitution / production (finite)
- $S \in V$, the start symbol.

If u, v, w are strings of variables & terminals

$A \to w$ then we say

uAv yields uwv written as

$$\underline{uAv} \Rightarrow \underline{uwv}.$$

derives

$u \overset{*}{\Rightarrow} v$ if $u = v$ or if a

sequence $u_1, u_2, \ldots, u_k$ exist $k \geq 0$
&

$$u \Rightarrow u_1 \Rightarrow u_2 \Rightarrow \ldots \Rightarrow u_k \Rightarrow v$$

$G = (\{S\}, \{a, b\}, R, S)$

$$\boxed{R = \{ \begin{array}{l} S \to aSb, \\ S \to SS, \\ S \to \varepsilon \end{array} \}}$$

$R = \{ S \to aSb \mid SS \mid \varepsilon \}$